

Equity Dilution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Pressure (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.54 (0.48), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.11.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic delays in how markets respond to value-relevant signals. We examine whether equity dilution pressure (EDP), measured through changes in shares outstanding relative to firm assets, predicts future stock returns due to gradual information diffusion. This investigation addresses a fundamental gap in understanding how corporate financing decisions transmit through equity markets when investors face attention constraints and information processing frictions.

The slow diffusion of information provides a compelling framework for understanding why equity dilution pressure predicts returns. When firms issue new equity or repurchase shares, these actions contain information about management's assessment of firm value and future prospects that diffuses gradually through the market [Hong and Stein \(1999\)](#). Limited investor attention, documented extensively in [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), creates systematic underreaction to dilution events as investors fail to immediately process the implications of changing share counts. This gradual incorporation mechanism predicts that firms experiencing high dilution pressure will see continued negative returns as the market slowly adjusts to the information content of equity issuance decisions.

The theoretical foundation for delayed price adjustment to financing activities builds on models of information diffusion with heterogeneous investors. Following [Hong et al. \(2000a\)](#), when information travels slowly across investor segments, prices initially underreact to news before gradually converging to fundamental values. For equity dilution signals, this process becomes particularly pronounced because share issuance information often arrives through regulatory filings and financial statements

that require active monitoring rather than through salient media coverage [Hirshleifer et al. \(2011\)](#). The complexity of interpreting dilution’s implications for firm value further slows the incorporation process, as investors must disentangle whether issuance reflects overvaluation, investment needs, or financial distress.

Cross-sectional variation in information frictions generates testable predictions about where dilution pressure effects should concentrate. Firms with lower analyst coverage experience slower information diffusion, as documented in [Hong et al. \(2000b\)](#), suggesting stronger return predictability from EDP among neglected stocks. Similarly, smaller firms face greater attention constraints and higher information acquisition costs, consistent with [Merton \(1987\)](#), implying that dilution pressure should generate larger return differentials where market frictions are most severe. These predictions align with broader evidence on post-earnings announcement drift and momentum, where return continuation concentrates among stocks with the highest barriers to information flow [Zhang \(2006\)](#).

The empirical evidence strongly supports the hypothesis that equity dilution pressure predicts returns through gradual information incorporation. A value-weighted long-short strategy based on EDP achieves an annualized gross Sharpe ratio of 0.54, with monthly average returns of 34 basis points (t -statistic = 4.25). The strategy’s alpha relative to the Fama-French six-factor model equals 22 basis points per month (t -statistic = 2.74), demonstrating that EDP captures a distinct return pattern beyond established risk factors. These results prove robust across alternative portfolio construction methodologies, with twenty-three of twenty-five specifications generating alphas with t -statistics exceeding two.

Consistent with information diffusion mechanisms, the predictive power of EDP concentrates where attention constraints bind most tightly. Among the largest quintile of stocks, where information flows most freely, the EDP strategy generates average returns of 19 basis points per month (t -statistic = 1.99), compared to 61 basis

points among the smallest quintile (t -statistic = 6.39). The strategy’s alpha relative to the six-factor model similarly declines from 42 basis points for small stocks to 12 basis points for large stocks, though statistical significance weakens substantially in the large-cap segment. This size-based variation in return predictability aligns precisely with theoretical predictions about how information frictions moderate the speed of price discovery.

Transaction cost analysis reveals that the economic magnitude of dilution pressure effects remains substantial after implementation frictions. The net Sharpe ratio of 0.48 ranks in the 98th percentile among 212 documented anomalies, while net returns of 30 basis points per month (t -statistic = 3.79) confirm the strategy’s implementability. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, EDP retains an alpha of 16 basis points per month (t -statistic = 2.11), establishing its incremental contribution to the cross-section of expected returns. The strategy’s strongest factor loading appears on CMA (0.33, t -statistic = 5.90), consistent with dilution pressure capturing investment-related information that propagates slowly through prices.

This paper contributes to three streams of literature by documenting how equity dilution pressure generates predictable returns through information diffusion channels. Relative to [Pontiff and Woodgate \(2008\)](#) and [Daniel et al. \(2006\)](#) in the *Journal of Finance*, who examine share issuance over one- and five-year horizons, we demonstrate that higher-frequency dilution pressure measures capture distinct information content that predicts returns incrementally. Our findings extend [Lyandres et al. \(2019\)](#) in the *Journal of Financial Economics* by showing that dilution effects persist after controlling for net payout yield, suggesting that gross dilution activity contains information beyond net financing flows. The methodological innovation of measuring dilution pressure through continuous changes in shares outstanding relative to assets provides a more granular signal than the discrete issuance indicators

employed in prior work.

Our evidence on cross-sectional variation in dilution pressure effects advances the literature on limited attention and gradual information diffusion. Building on [Cohen and Frazzini \(2008\)](#) in the *Journal of Finance*, who document slow diffusion of information across economically linked firms, we show that information about a firm’s own financing activities also propagates gradually when attention constraints bind. The concentration of return predictability among smaller, less-analyzed firms extends [Hou et al. \(2009\)](#) in the *Review of Financial Studies* by demonstrating that market frictions impede the incorporation of accounting-based signals beyond earnings. Our results complement [Richardson et al. \(2005\)](#) in the *Journal of Accounting Research* by isolating the equity issuance component of accruals and showing its distinct role in return prediction through information channels.

The broader implications extend to understanding market efficiency and the limits of arbitrage. The persistence of dilution pressure effects, even after accounting for transaction costs and controlling for 212 anomalies from the factor zoo, suggests that information processing constraints create economically meaningful inefficiencies. The finding that EDP improves the mean-variance efficient frontier for 98% of net performance comparisons demonstrates that gradual information diffusion generates genuine alpha rather than compensation for systematic risk. These results inform the debate on anomaly publication effects examined in [McLean and Pontiff \(2016\)](#) in the *Journal of Finance*, as the economic magnitude of dilution pressure remains substantial despite the widespread availability of share issuance data, highlighting the role of interpretation complexity in sustaining return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. construction of our Equity Dilution Pressure signal requires two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common equity and increases when firms issue new shares or conduct stock-based transactions. Current assets (ACT) captures the total value of assets expected to be converted to cash or consumed within one year or the operating cycle, including cash, marketable securities, accounts receivable, and inventory. This measure provides a natural scaling variable that reflects the firm’s short-term resource base. construct the Equity Dilution Pressure signal by computing the year-over-year change in common stock scaled by lagged current assets: $(CSTK(t) - CSTK(t-1)) / ACT(t-1)$. This ratio captures the magnitude of new equity issuance activity relative to the firm’s liquid asset base. Positive values indicate increases in common stock, potentially signaling equity dilution through new share issuances, while negative values may reflect share buybacks or retirements. We calculate this measure using fiscal year-end values from annual financial statements and require non-missing values for both CSTK and ACT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.07 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.04 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 5.49% of CRSP names, which on average make up 6.59% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.34% per month with a t-statistic of 4.25. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.33,

with a t-statistic of 5.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 528 stocks and an average market capitalization of at least \$1,171 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 19-36bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 19 bps/month with a t-statistic of 1.99. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 12 to 17 bps/month with t-statistics between 1.22 and 1.79.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.54 (0.48) is greater than 92% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$8.80 which ranks the EDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$6.48 which ranks the EDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 2.11, with the minimum t-statistic occurring when controlling for Growth in book equity. Controlling for all six closely related anomalies, the t-statistic on EDP is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Growth in book equity. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.11.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDP signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDP grows to \$3098.64.

8 Conclusion

This study provides compelling evidence that Equity Dilution Pressure (EDP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized Sharpe ratio of 0.54 gross (0.48 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22 bps/month gross, t-statistic = 2.74) suggests that EDP captures a distinct dimension of expected returns not fully explained by conventional risk factors. The persistence of significant alpha (16 bps/month, t-statistic = 2.11) even after controlling for six closely related equity issuance and growth-based strategies underscores the unique information content embedded in the EDP signal.

ization on CRSP in the period for which EDP is available.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The signal’s robustness to transaction costs and its orthogonality to existing factors suggest that EDP could serve as a valuable addition to multi-factor investment strategies. The economic magnitude of the returns, combined with statistical significance, indicates that markets may systematically misprice firms experiencing dilution pressure, potentially due to investor inattention to the complex implications of equity financing decisions.

Several limitations of this study warrant consideration. First, while we employ state-of-the-art methodologies to address data-snooping concerns, the true out-of-sample performance of EDP remains to be validated through extended real-time implementation. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, the underlying economic mechanisms driving the EDP anomaly deserve deeper theoretical and empirical exploration.

Future research should pursue several promising directions. Investigating the time-varying nature of the EDP premium and its relationship to macroeconomic conditions could provide insights into when the strategy performs best. Additionally, examining the interaction between EDP and corporate governance characteristics might reveal whether the anomaly is more pronounced in firms with weaker monitoring mechanisms. Finally, exploring the behavioral underpinnings of investor responses to equity dilution events could enhance our understanding of why this pricing inefficiency persists. As markets continue to evolve and new data becomes available, continued scrutiny of the EDP signal will be essential to confirm its lasting relevance as a return predictor.

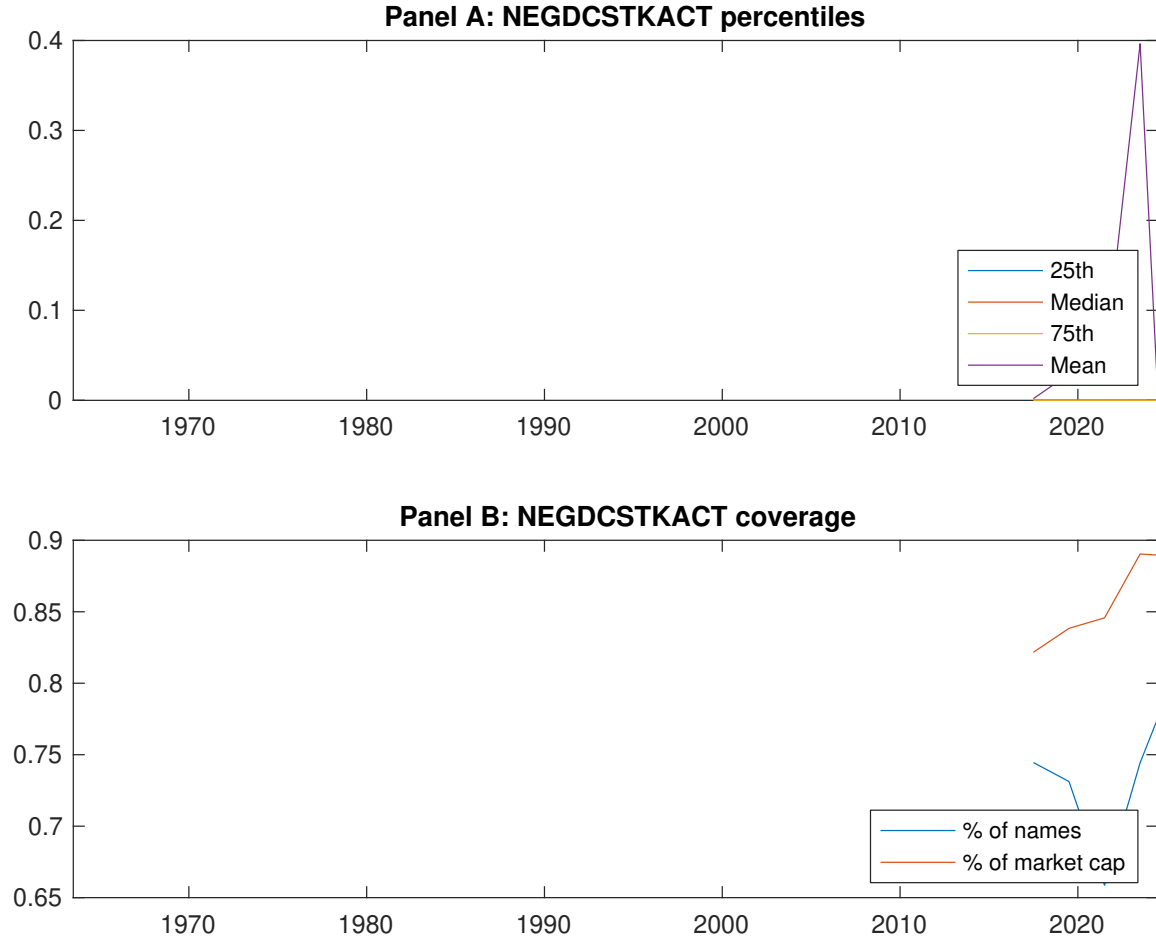


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.80]	0.51 [2.73]	0.64 [3.60]	0.65 [4.03]	0.79 [4.97]	0.34 [4.25]
α_{CAPM}	-0.09 [-1.67]	-0.12 [-2.19]	0.04 [0.69]	0.10 [2.06]	0.25 [4.95]	0.34 [4.33]
α_{FF3}	-0.08 [-1.47]	-0.07 [-1.27]	0.09 [1.83]	0.10 [2.01]	0.24 [4.63]	0.32 [3.98]
α_{FF4}	-0.07 [-1.33]	-0.04 [-0.69]	0.09 [1.81]	0.06 [1.25]	0.21 [4.02]	0.28 [3.49]
α_{FF5}	-0.11 [-1.92]	0.01 [0.21]	0.11 [2.09]	0.01 [0.27]	0.14 [2.73]	0.24 [3.02]
α_{FF6}	-0.10 [-1.78]	0.03 [0.57]	0.11 [2.08]	-0.01 [-0.20]	0.12 [2.43]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	0.94 [69.20]	1.01 [79.17]	0.99 [78.05]	0.98 [83.03]	0.96 [79.79]	0.03 [1.49]
β_{SMB}	-0.03 [-1.52]	0.04 [2.27]	0.09 [5.01]	-0.05 [-2.98]	0.01 [0.54]	0.04 [1.38]
β_{HML}	-0.00 [-0.13]	-0.13 [-5.41]	-0.20 [-8.28]	-0.07 [-3.23]	-0.07 [-3.12]	-0.07 [-1.85]
β_{RMW}	0.10 [3.76]	-0.15 [-6.15]	-0.07 [-2.68]	0.10 [4.31]	0.13 [5.62]	0.03 [0.88]
β_{CMA}	-0.04 [-1.06]	-0.11 [-3.05]	0.05 [1.37]	0.25 [7.54]	0.29 [8.33]	0.33 [5.90]
β_{UMD}	-0.01 [-0.69]	-0.03 [-2.15]	-0.00 [-0.10]	0.03 [2.81]	0.02 [1.59]	0.03 [1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	578	617	528	579	629	
me (\$10 ⁶)	1487	1171	1740	1882	2203	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.25]	0.34 [4.33]	0.32 [3.98]	0.28 [3.49]	0.24 [3.02]	0.22 [2.74]
Quintile	NYSE	EW	0.56 [7.97]	0.63 [9.02]	0.55 [8.30]	0.47 [7.16]	0.40 [6.35]	0.35 [5.53]
Quintile	Name	VW	0.33 [4.20]	0.34 [4.33]	0.31 [3.99]	0.29 [3.66]	0.25 [3.16]	0.24 [3.00]
Quintile	Cap	VW	0.23 [2.87]	0.23 [2.82]	0.22 [2.68]	0.18 [2.16]	0.18 [2.21]	0.15 [1.85]
Decile	NYSE	VW	0.33 [3.43]	0.29 [3.02]	0.27 [2.75]	0.21 [2.14]	0.22 [2.24]	0.18 [1.81]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.79]	0.30 [3.85]	0.28 [3.49]	0.26 [3.24]	0.22 [2.79]	0.21 [2.64]
Quintile	NYSE	EW	0.36 [4.73]	0.45 [5.89]	0.37 [5.14]	0.33 [4.63]	0.22 [3.33]	0.20 [3.02]
Quintile	Name	VW	0.29 [3.73]	0.30 [3.82]	0.27 [3.48]	0.26 [3.31]	0.22 [2.86]	0.22 [2.78]
Quintile	Cap	VW	0.19 [2.43]	0.19 [2.37]	0.18 [2.21]	0.16 [1.93]	0.15 [1.92]	0.14 [1.73]
Decile	NYSE	VW	0.29 [3.02]	0.25 [2.57]	0.23 [2.33]	0.19 [2.01]	0.19 [1.93]	0.17 [1.70]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.36]	0.70 [2.60]	0.91 [3.34]	0.95 [3.79]	0.95 [4.05]	0.61 [6.39]	0.67 [7.11]	0.60 [6.68]	0.55 [6.08]	0.45 [5.11]	0.42 [4.72]
	(2)	0.58 [2.46]	0.69 [2.82]	0.84 [3.43]	0.96 [4.12]	0.90 [4.09]	0.32 [3.63]	0.38 [4.36]	0.28 [3.45]	0.25 [2.96]	0.20 [2.45]	0.18 [2.15]
	(3)	0.56 [2.78]	0.63 [2.80]	0.77 [3.38]	0.82 [3.92]	0.92 [4.63]	0.36 [4.26]	0.37 [4.39]	0.31 [3.71]	0.30 [3.47]	0.26 [3.08]	0.26 [2.95]
	(4)	0.49 [2.68]	0.63 [3.08]	0.78 [3.76]	0.82 [4.27]	0.78 [4.29]	0.29 [3.54]	0.30 [3.63]	0.23 [2.90]	0.24 [2.97]	0.05 [0.68]	0.08 [1.01]
	(5)	0.51 [3.29]	0.47 [2.61]	0.53 [2.98]	0.54 [3.26]	0.70 [4.50]	0.19 [1.99]	0.17 [1.79]	0.17 [1.74]	0.14 [1.39]	0.14 [1.45]	0.12 [1.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	330	331	331	328	324	24	27	33	25	23	
	(2)	92	90	91	91	89	47	46	47	47	46	
	(3)	66	65	65	65	65	81	78	80	81	83	
	(4)	56	56	56	56	56	176	174	182	181	185	
(5)	51	51	51	51	51	1274	1146	1373	1408	1619		

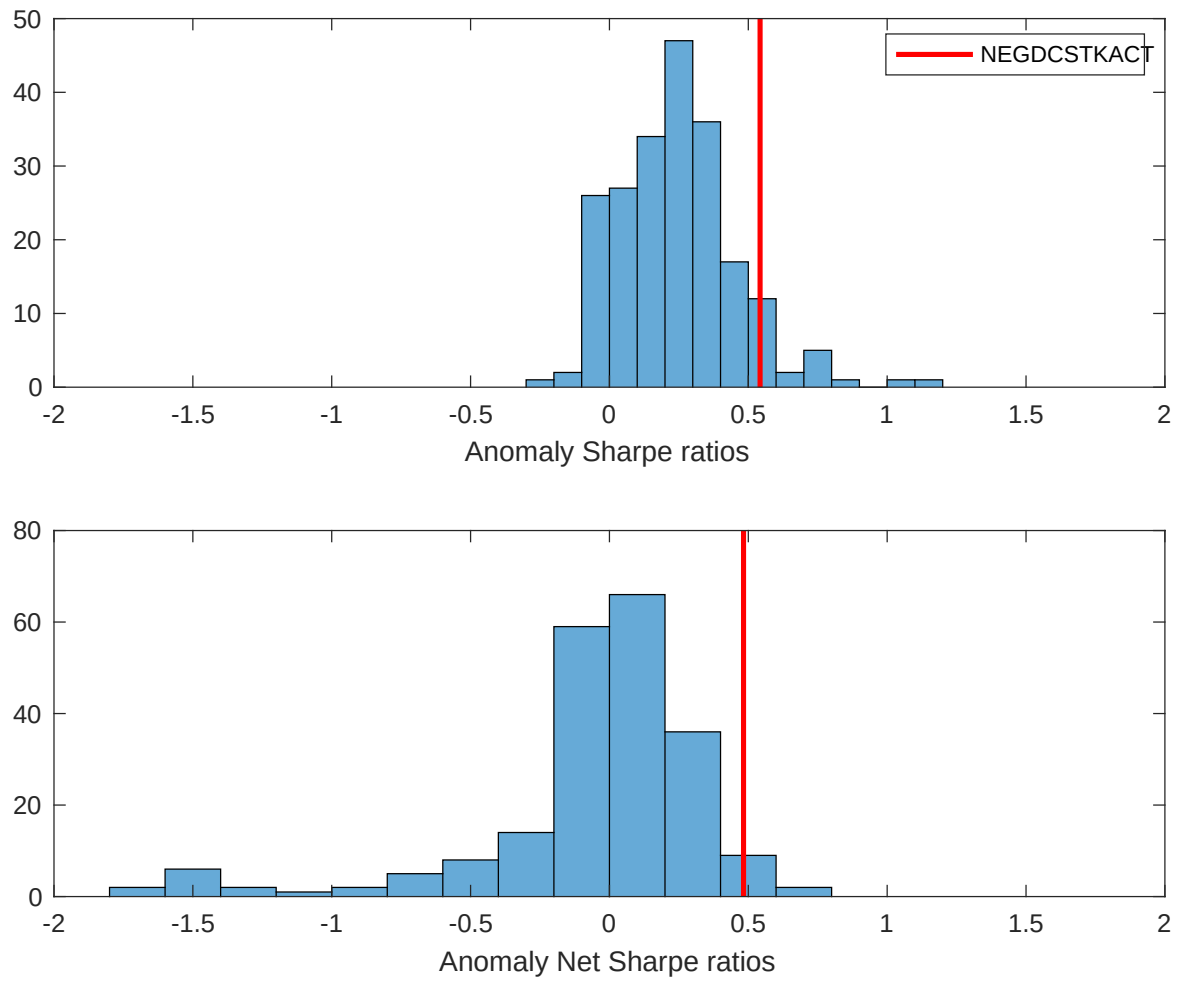


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

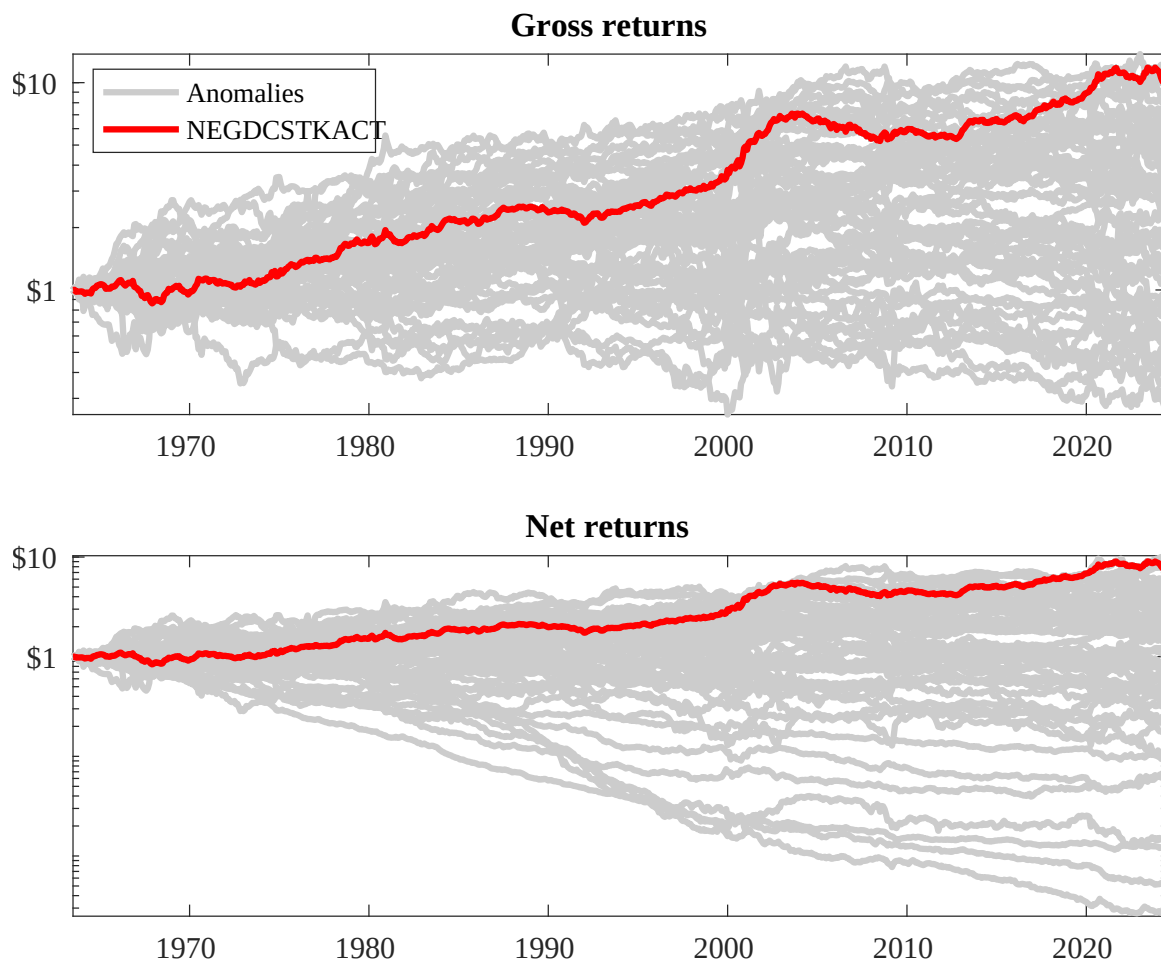
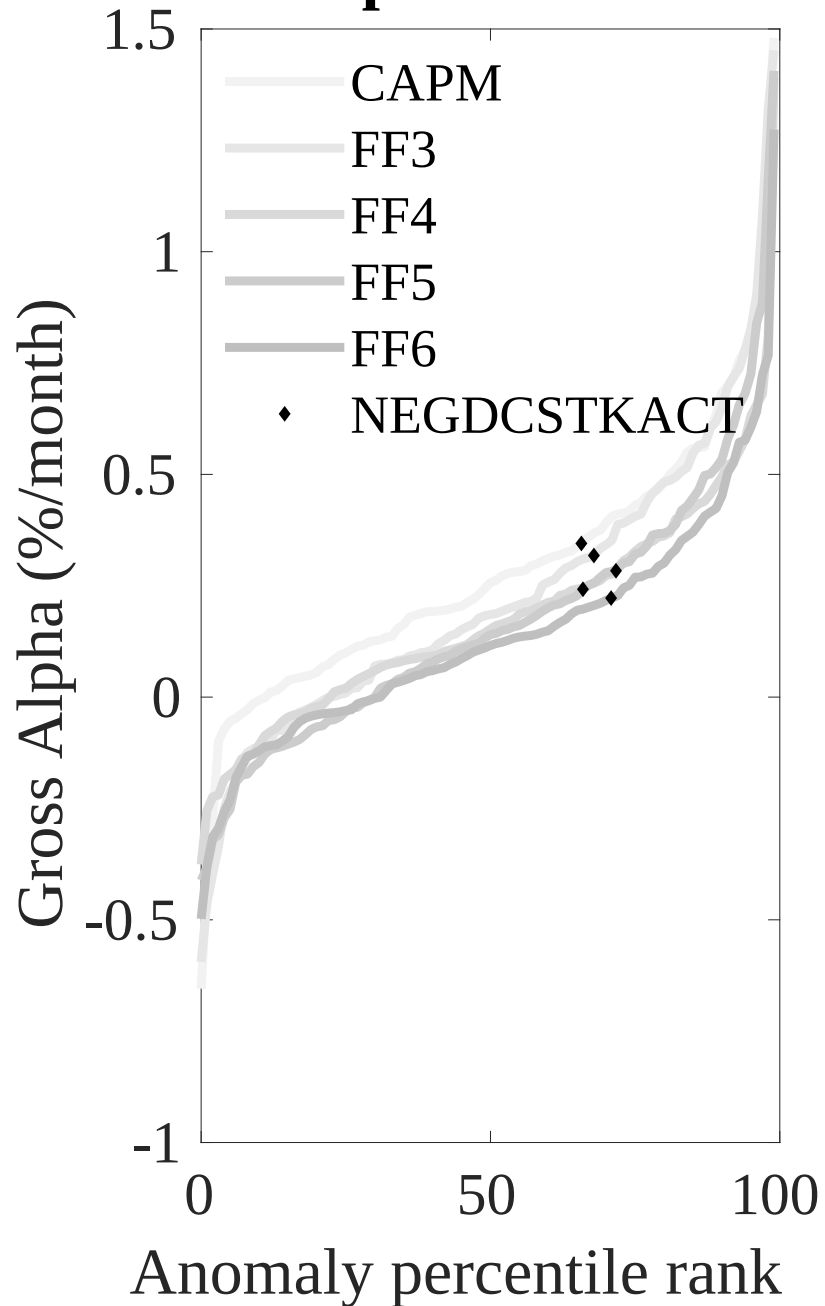


Figure 3: Dollar invested.

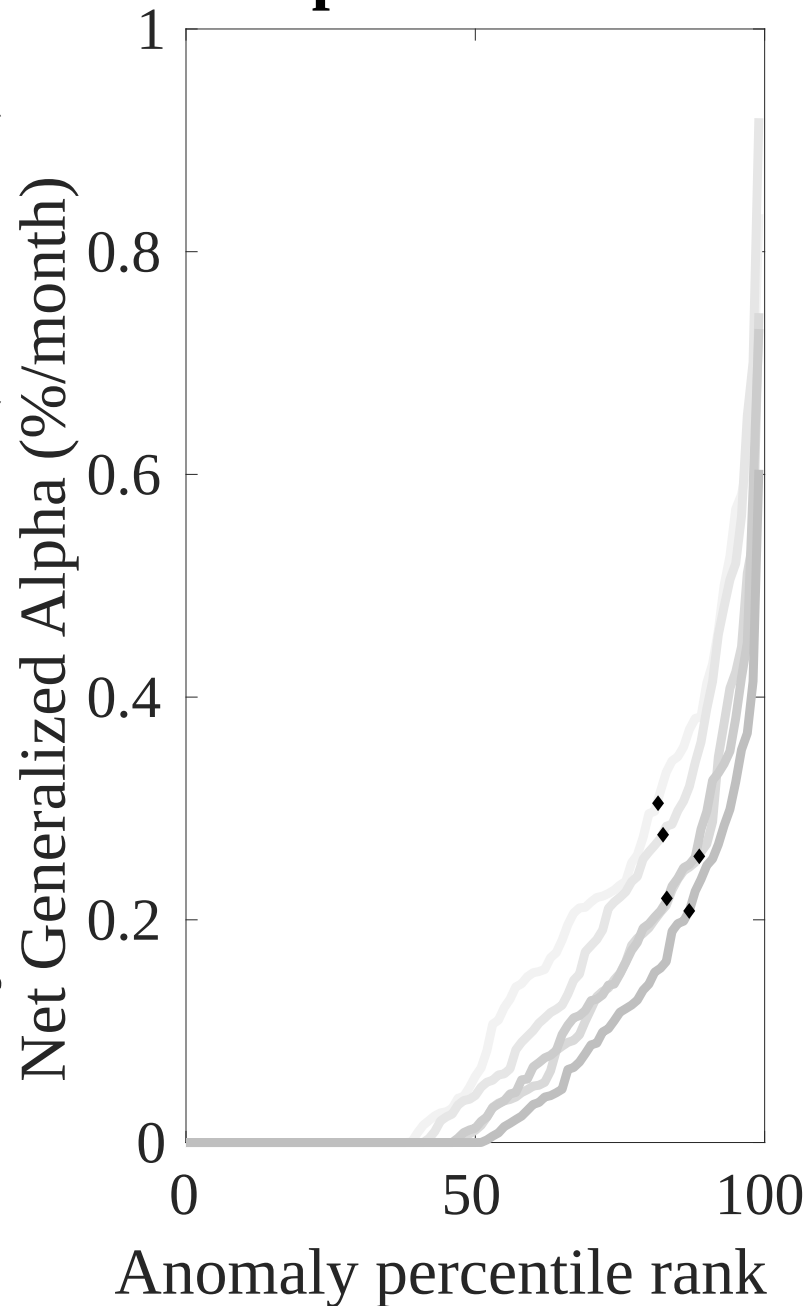
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



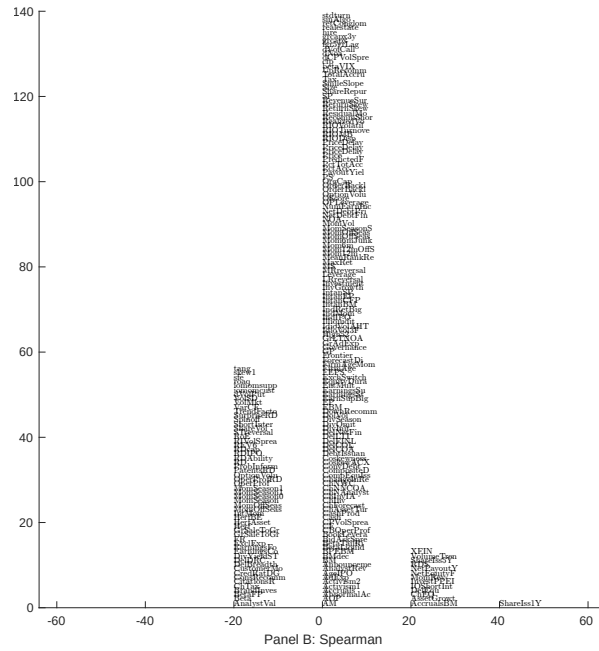
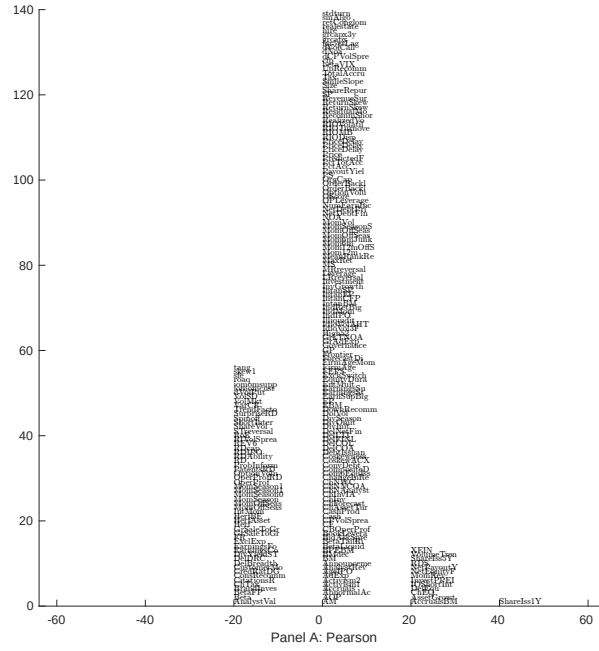


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

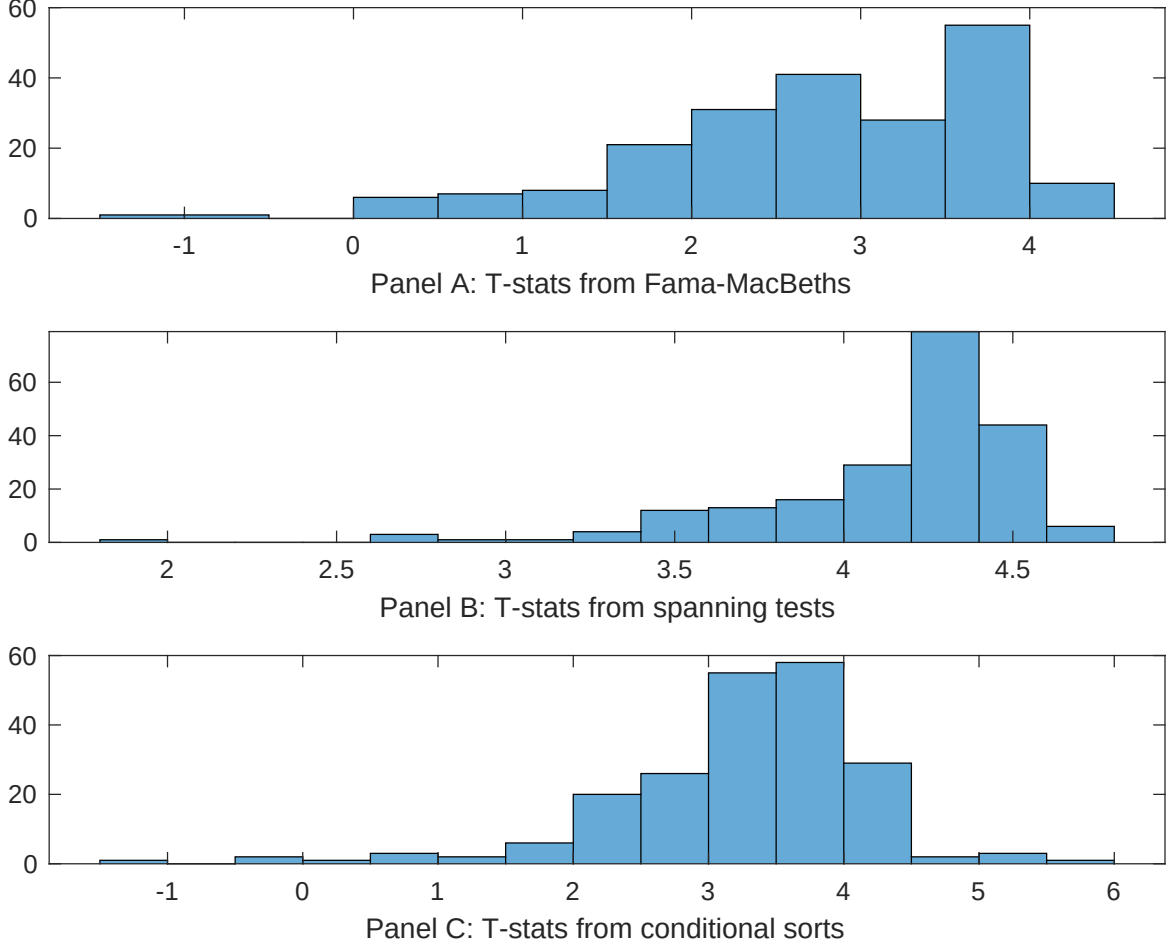


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.13 [6.30]	0.17 [6.71]	0.12 [5.39]	0.13 [5.51]	0.13 [5.82]	0.11 [3.68]
EDP	0.14 [2.42]	0.12 [2.46]	0.14 [2.11]	0.12 [2.12]	0.18 [2.54]	0.15 [2.24]	0.77 [1.47]
Anomaly 1	0.14 [5.85]						0.59 [2.16]
Anomaly 2		0.27 [4.87]					0.25 [4.33]
Anomaly 3			0.39 [3.21]				-0.18 [-0.75]
Anomaly 4				0.21 [1.84]			0.10 [0.98]
Anomaly 5					0.12 [3.28]		0.30 [0.48]
Anomaly 6						0.13 [9.32]	0.83 [5.96]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.56]	0.15 [1.85]	0.20 [2.61]	0.25 [3.09]	0.22 [2.73]	0.20 [2.50]	0.16 [2.11]
Anomaly 1	29.42 [7.42]						13.47 [2.79]
Anomaly 2		28.62 [6.81]					14.35 [2.99]
Anomaly 3			31.52 [7.45]				30.12 [4.96]
Anomaly 4				16.03 [5.19]			4.67 [1.38]
Anomaly 5					14.56 [3.65]		-11.13 [-1.98]
Anomaly 6						9.80 [2.16]	3.36 [0.76]
mkt	3.76 [2.01]	4.32 [2.30]	4.08 [2.19]	5.53 [2.83]	2.78 [1.45]	3.10 [1.61]	5.94 [3.14]
smb	6.24 [2.31]	6.98 [2.56]	4.01 [1.49]	7.66 [2.74]	4.88 [1.76]	5.17 [1.86]	6.57 [2.43]
hml	-6.53 [-1.81]	-3.44 [-0.95]	-9.61 [-2.64]	-11.93 [-3.07]	-7.65 [-2.05]	-6.47 [-1.73]	-9.37 [-2.49]
rmw	-12.48 [-2.95]	-6.80 [-1.72]	4.95 [1.36]	-5.87 [-1.42]	4.79 [1.28]	3.53 [0.94]	-11.20 [-2.50]
cma	13.62 [2.34]	18.00 [3.17]	1.33 [0.20]	18.35 [3.08]	17.21 [2.55]	24.55 [3.82]	-7.90 [-1.09]
umd	1.59 [0.86]	1.07 [0.57]	2.39 [1.29]	4.22 [2.23]	3.30 [1.74]	2.76 [1.44]	0.86 [0.46]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	12	11	12	9	7	6	17

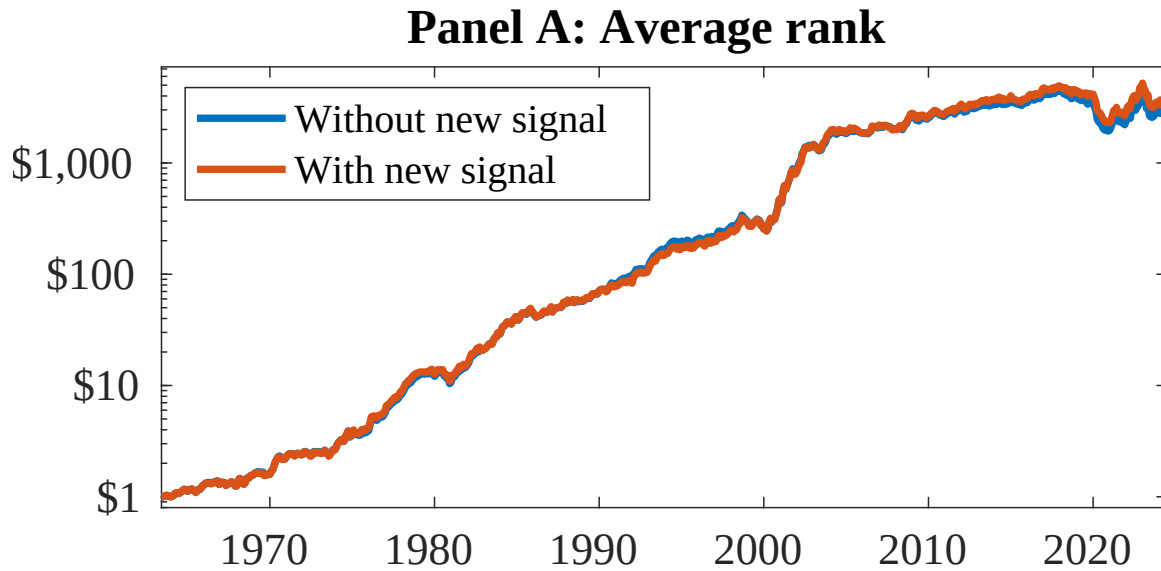


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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